UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Friday 9 June 2023 13.30 to 16.30 (*3 hours plus any pre-agreed individual adjustment*)

Paper PHA/WP

PHARMACOLOGY WRITTEN PRACTICAL PAPER

*Answer **all four** questions. Each **question** is worth **20 marks in total**.*

Extended Answer questions (2) and (4) have no word limit

Diagrams or other handwritten material
Where necessary to perform, show your mathematical workings.
Where appropriate (e.g., for formulae, equations, diagrams or mathematical calculations) you may hand write your answers/workings using the Inspera Scan Paper provided. You should not copy and paste images from any other documents.

Please Note
- *Inspera Scan question codes MUST be noted during the exam.*
- *Please ensure that each Inspera Scan page is labelled appropriately, including your Blind Grade Number.*
- *Inspera Scan pages MUST be placed in the Inspera Scan folder at the completion of the exam and left on your desk.*
- *Inspera scan pages will be scanned by administrators after the completion of the exam.*

Stationery requirements
- *Rough work pad*
- *Inspera Scan pages*
- *Inspera Scan Folder*

Calculator – students are permitted to use an approved calculator

Question 1

Stimulation of adenosine A_3 receptors (A_3Rs) has been implicated in the mechanism of several important responses to adenosine, including vasodilatation, bronchoconstriction and cardioprotection. These receptors have also been found to be overexpressed in some tumour and immune cells which has positioned them as attractive anti-cancer and anti-inflammatory targets in recent times. CB2 Pharmaeuticals recently developed a novel, adenosine-based selective agonist of human A_3Rs (hA_3Rs) namely CB2-MECA (**Figure 1**) cardioprotection. However, it is not known what effect prolonged treatment with CB2-MECA has on A_3Rs function. The following studies were therefore designed to examine the effects of prolonged exposure of Chinese Hamster Ovary (CHO) cells transfected with recombinant hA_3Rs to CB2-MECA. CB2 Pharmaceuticals also developed a radioactive version of the same drug namely [^{3}H] CB2-MECA which does not significantly differ from CB2-MECA in its binding affinity for hA_3R.

In the first set of experiments, binding of [^{3}H] CB2-MECA to non-tranfsected CHO cells and those transfected with hA_3Rs was measured using various concentration of [^{3}H] CB2-MECA in the absence and presence of 1 μM unlabelled CB2-MECA. The results are shown in **Table 1** and **Figure 2**.

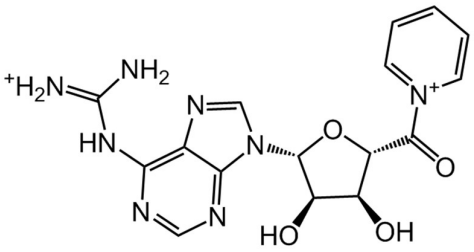

Figure 1. 2D chemical structure of CB2-MECA

Table 1 binding of [^{3}H] CB2-MECA in transfected and non-transfected CHO cells.

	[^{3}H]CB2-MECA binding (**B**, pmol/mg protein)			
	CHO cells transfected with recombinant hA_3Rs		non-transfected CHO cells	
Concentration of [^{3}H] CB2-MECA (**D**, nM)	no unlabelled CB2-MECA	with unlabelled CB2-MECA	no unlabelled CB2-MECA	with unlabelled CB2-MECA
1	3.2	0.2	0.1	0.2
2	4.9	0.4	0.5	0.4
4	6.8	0.8	0.7	0.8
8	8.8	1.6	1.7	1.6
30	14.4	6.0	6.2	6.0

Note: 1 pmol = 10^{-12} mol; 1 fmol = 10^{-15} mol

[Question 1 continues on the NEXT PAGE]

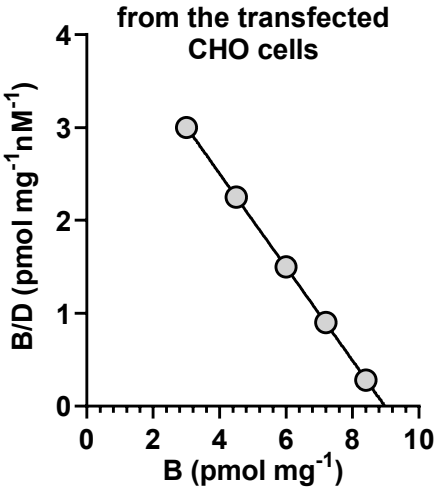

***Figure 2**. The Scatchard plot relevant to specific binding of [^{3}H]-CB2-MECA in CHO cells transfected with hA$_3$Rs. Data contained in this plot are in Table 1.*

A. Determine the concentration of CB2-MECA that would be required to occupy half of the hA$_3$Rs expressed in transfected CHO cells and provide a rationale to your answer (3 marks). What conditions need to be fulfilled for your calculations to be accurate? (3 marks).

B. Calculate the maximum number of hA$_3$Rs per gram of total protein present in transfected CHO cells (Avogadro's number = 6.02 x 10^{23}) (**3 marks**).

C. Comment on the results obtained with non-transfected cells (1 **mark)**.

In a further series of experiments, CHO cells transfected with hA$_3$Rs were exposed to 1 μM of CB2-MECA for 1 hour, after which the buffer containing CB2-MECA was replaced by fresh buffer devoid of any CB2-MECA. Measurements were then made of the inhibition of [^{3}H] CB2-MECA (1 nM) binding by various concentrations of unlabelled CB2-MECA. In these experiments, the specific binding of 1 nM [^{3}H] CB2-MECA in the absence of unlabelled CB2-MECA was found to be 17 fmol/mg protein. Furthermore, the concentration of unlabelled CB2-MECA that inhibited the specific [^{3}H] CB2-MECA binding by 50% was 160 nM.

D. Did pretreatment of CHO cells expressing hA$_3$Rs with CB2-MECA affect the ability of these receptors to bind this drug? If so, what can you deduce from these results as the possible underlying reasons? For both these questions, use calculations to provide the answers (**2+4 = 6 marks**).

A$_3$Rs activation inhibits cyclic AMP (cAMP) production induced by direct stimulation of adenylyl cyclase with a natural product called forskolin. In the next series of experiments, the ability of CB2-MECA to inhibit forskolin-induced cAMP production was tested in CHO cells transfected with hA$_3$Rs before (No pretreatment), and after (With pretreatment) exposure to 1 μM CB2-MECA for 1 hour. The data are shown in **Table 2** and **Figure 3**.

[Question 1 continues on the NEXT PAGE]

Table 2 Effect of graded concentrations of CB2-MECA on inhibition of cAMP production in CHO cells expressing hA_3Rs with and without prior treatment with the same drug. Each data point represents the mean value obtained for ≥100 cells from experiments done in triplicate.

No pretreatment		With pretreatment	
Concentration of CB2-MECA (M)	% inhibition of cAMP production	Concentration of CB2-MECA (M)	% inhibition of cAMP production
$1x10^{-13}$	3	$1x10^{-9}$	4
$3x10^{-13}$	10	$1x10^{-8}$	11
$1x10^{-12}$	25	$3x10^{-8}$	25
$3x10^{-12}$	46	$1x10^{-7}$	48
$1x10^{-11}$	65	$3x10^{-7}$	66
$3x10^{-11}$	75	$1x10^{-6}$	75
$1x10^{-10}$	78	$3x10^{-6}$	78
$3x10^{-10}$	80	$1x10^{-5}$	80

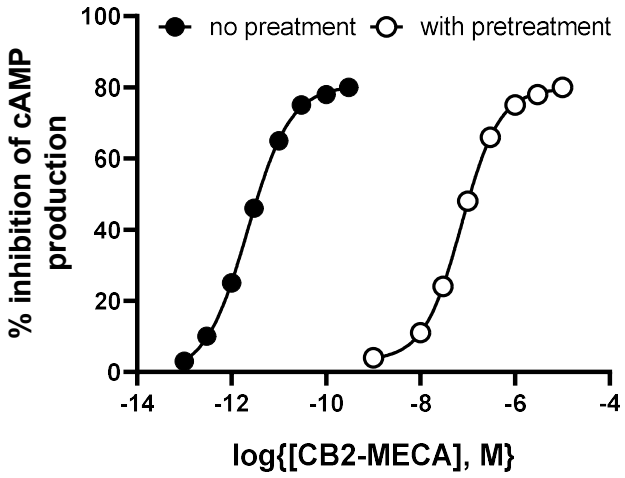

***Figure 3**. Effect of various concentrations of CB2-MECA on the inhibition of forskolin-induced cAMP production in CHO cells expressing hA_3Rs with and without prior treatment with 1μM of CB2-MECA for 1hr. Each data point represents mean of ≥100 cells (no error bar shown) from experiments done in triplicate.*

E. **Comment on the results shown in Table 2 and Figure 3** in the context of your calculations and conclusions for questions A to D (**4 marks**).

End of Question 1

2. What determines whether a drug binds to and activates a particular receptor? Give reference to specific drugs in your answer. (20 marks)

[Question 3 starts on the NEXT PAGE]

Question 3

Cantabutamol is a novel, short-acting β2 adrenoceptor selective agonist developed by CamPharm Biotech for primarily preventing premature labour but the drug can also be used in acute and chronic asthma. Like many other sympathomimetic amines, it has a chiral centre and exists as two enantiomers namely (+) and (-) Cantabutamol but only the (-) enantiomer is able to relax bronchial or uterine smooth muscle. Although it was challenging, the chemists at the CB2 Pharmaceuticals were able to obtain both the enantiomers in optically pure form but their main product named Cantab™ is a racemic mixture (1:1) of (+) and (-) Cantabutamol.

In Phase I clinical trial, 20 healthy female volunteers were each given 3 mg of either (+) or (-) Cantabutamol by intravenous injection and the plasma concentrations of the drug were then determined at various time points after the injection. The results are shown **Figure 1** and the corresponding values are given in **Table 1**.

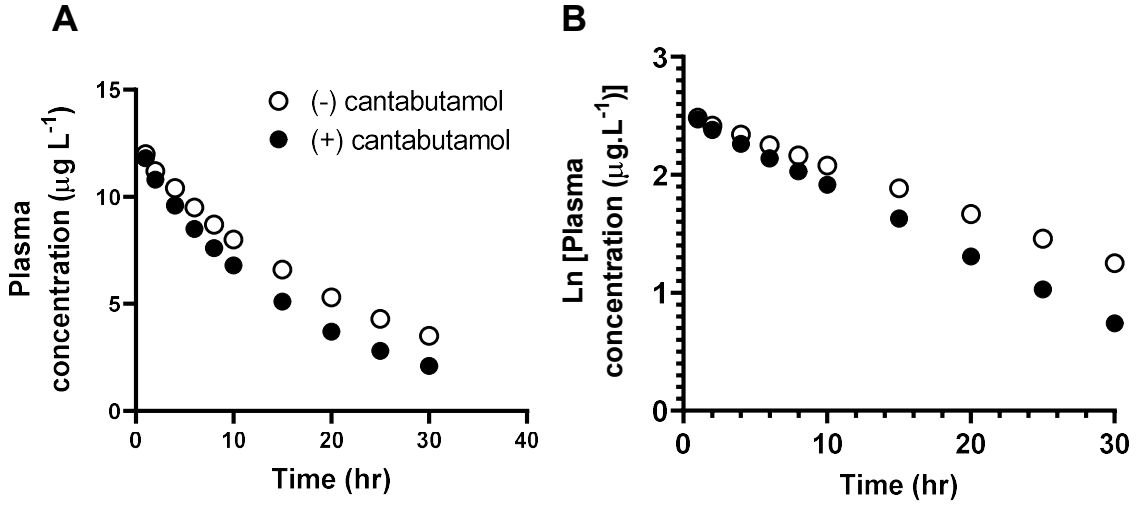

Figure 1. *Plasma concentration versus time plot of the Cantabutamol enantiomers. Each data point represents average plasma concentration (μg L^{-1}, error bar not show) of the drug from 10 female volunteers receiving either the (+) (shown in filled circle) or (-) enantiomer.* ***B*** *is the semi-logarithmic version of the linear plot shown in* ***A****. The data for the plots are also given in Table 1.*

A. Determine the half-life, clearance and volume of distribution for the two enantiomers of Cantabutamol. (3 marks)

[Question 3 continues on the NEXT PAGE]

Table 1. **Plasma concentrations of Cantabutamol enantiomers determined at various times after i.v. injection**

Time (hr)	Plasma concentration ($\mu g.L^{-1}$)	
	(-) Cantabutamol	**(+) Cantabutamol**
1	12.0	11.8
2	11.2	10.8
4	10.4	9.6
6	9.5	8.5
8	8.7	7.6
10	8.0	6.8
15	6.6	5.1
20	5.3	3.7
25	4.3	2.8
30	3.5	2.1

The volunteers voided their urine completely 20 hours after receiving the injection of the drug, and a urine sample was collected for each of them later at the end of 25th hour. The average volume of urine in the samples was 300 ml for both groups, and the concentrations of the drug in the urine were 278 $\mu g\ L^{-1}$ for the (-) Cantabutamol recipient group and 406 $\mu g\ L^{-1}$ for the (+) Cantabutamol recipient group.

B. Assuming that the binding of either enantiomers to plasma proteins was negligible, calculate the renal clearance of the two enantiomers (2 marks). Comment on the results (**4 marks**).

C. Two weeks after the first trial, each of the same 20 female volunteers were given Cantab™ as a single oral dose of 6mg. From the corresponding plasma concentration versus time data, it was found that the plasma and renal clearances of Cantab™ were not significantly different from the corresponding values for the (+) enantiomer obtained in earlier trial with i.v. infusion. **Comment on these results.** (**3 marks**)

[Question 3 continues on the NEXT PAGE]

D. The minimum effective concentration (MEC) of the bioactive Cantabutamol is 2.3 $\mu g\ L^{-1}$. **What infusion rate would be needed to achieve this concentration, if this enantiomer were given by constant i.v. infusion? How long would it take to reach 90% of the MEC?** (**1+1= 2 marks**)

E. What infusion rate would be needed to reach the MEC of the bioactive Cantabutamol in 1 hr? (2 marks)

F. What infusion rate needs to be used for Cantab™ for a patient undergoing premature labour to start benefiting from this product in 1 hr? (4 marks)

End of Question 3

4. Compare and contrast the roles of enzymes and transporters in drug elimination. (20 marks)

END OF PAPER